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Paleomagnetism of the Permian Wallowa terrane: Implications for terrane migration and orogeny

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Abstract. A sequence of volcaniclastic sedimentary and pyroclastic rocks of Permian age (late Leonardian or early Guadalupian; ≈260 Ma) in the Wallowa terrane of Oregon and Idaho has been sampled for paleomagnetic study. Forty-six sites (222 cores) were collected from folded, laminated siltstone, mudstone, and tuffaceous sandstone beds exposed along the Snake River in Hells Canyon. Several stratigraphic sections, within the 1500- to 2500-m-thick Hunsaker Creek Formation, were sampled over a stratigraphic interval of about 700 m. Thermal demagnetization revealed three components of magnetization. The A component has the lowest unblocking temperatures (generally less than about 200°C), has normal polarity, and fails the fold test. The B component is removed between heating steps of about 200°C and 550°C, is of normal polarity, and also fails the fold test. The C component, observed between heating steps of approximately 550°C and 660°C, is always of negative inclination. The grouping of this component improves slightly upon unfolding, with the optimum grouping occurring at 95% unfolding, using sites with $n \geq 3$ and $k \geq 10$, ($k95 / k0$=1.18), or 100% using sites $n \geq 3$ ($k95 / k0$=1.51). This component is interpreted as the characteristic remanent magnetization acquired during deposition of the Hunsaker Creek Formation based on the following observations: The better grouping of the high-temperature magnetic component after unfolding and the consistent reversed polarity seen in samples from the different sections. The resulting paleolatitude, calculated using inclination-only statistics, of 24°± 12° N ($n \geq 3$, $k \geq 10$), or 26°± 9°N ($n \geq 3$ sites), implies a northern hemisphere position of the Wallowa terrane during the Early Permian and possible southward transport of this terrane with respect to North America. If the Wallowa terrane was attached to the Wrangellia superterrane, our result indicates a northern hemisphere location for that superterrane during the Permian.

Introduction

Hells Canyon contains excellent exposures of upper Paleozoic and lower Mesozoic strata, which hold clues to the tectonic assembly of the Blue Mountains province in eastern Oregon and western Idaho (Figure 1). These strata, the Seven Devils Group, are within the belt of suspect terranes, along the western margin of cratonic North America [*Hamilton*, 1963; *Brooks and Vallier*, 1978; *Hamilton*, 1978; *Davis et al.*, 1978]. As summarized by *Beck* [1989], paleomagnetic studies in the western Cordillera have favored the mobilistic paradigm of "microplate tectonics," in which translations and rotations of crustal blocks are commonplace. The blocks might consist of volcanic island arcs, oceanic plateaus, or continental fragments that either collided in a subduction zone or were displaced by transcurrent faulting along the North America plate margin. Paleomagnetic evidence has been collected in the Cordillera from Mexico to Alaska, showing that some terranes were merely "jostled," while other truly exotic terranes have moved a thousand kilometers or more. The Seven Devils Group is a prime candidate for study as a displaced terrane, owing to its island arc affinity and juxtaposition against the craton.

Hillhouse et al. [1982] reported a Triassic paleolatitude of 18° (north or south) for volcanogenic rocks in the Seven Devils Group. The ambiguity of hemisphere arose from not knowing the polarity of the geomagnetic field, recorded by the sediments and volcanic flows sampled in that study. From the Late Triassic cratonic reference pole, the predicted position of Hells Canyon in Late Triassic time would be 19°N, in keeping with the Seven Devils Group results. However, if the southern hemisphere is the correct placement of the Seven Devils paleolatitude, a northward displacement of 4000 km would be required to bring the terrane to its present position relative to North America [*Hillhouse and Gromme*, 1984].

Triassic rock of the Seven Devils Group yielded a magnetic overprint interpreted by *Hillhouse et al.* [1982] to be of Late Jurassic or Early Cretaceous age. The corresponding paleomagnetic pole of the overprint is significantly displaced from the North American reference pole, which led *Hillhouse et al.* [1982] to infer that the crustal block of the eastern Blue Mountains had rotated clockwise 66° about a vertical axis, following a similar interpretation for results of Late Jurassic plutons of the Blue Mountains [*Wilson and Cox*, 1980].

Our current study was designed to resolve the ambiguity regarding the hemisphere of origin of the Seven Devils Group and to test further the rotational hypothesis. The geomagnetic field maintained reversed polarity throughout all but a few brief intervals of Permian time [*Irving and Parry*, 1963; *Harland et*

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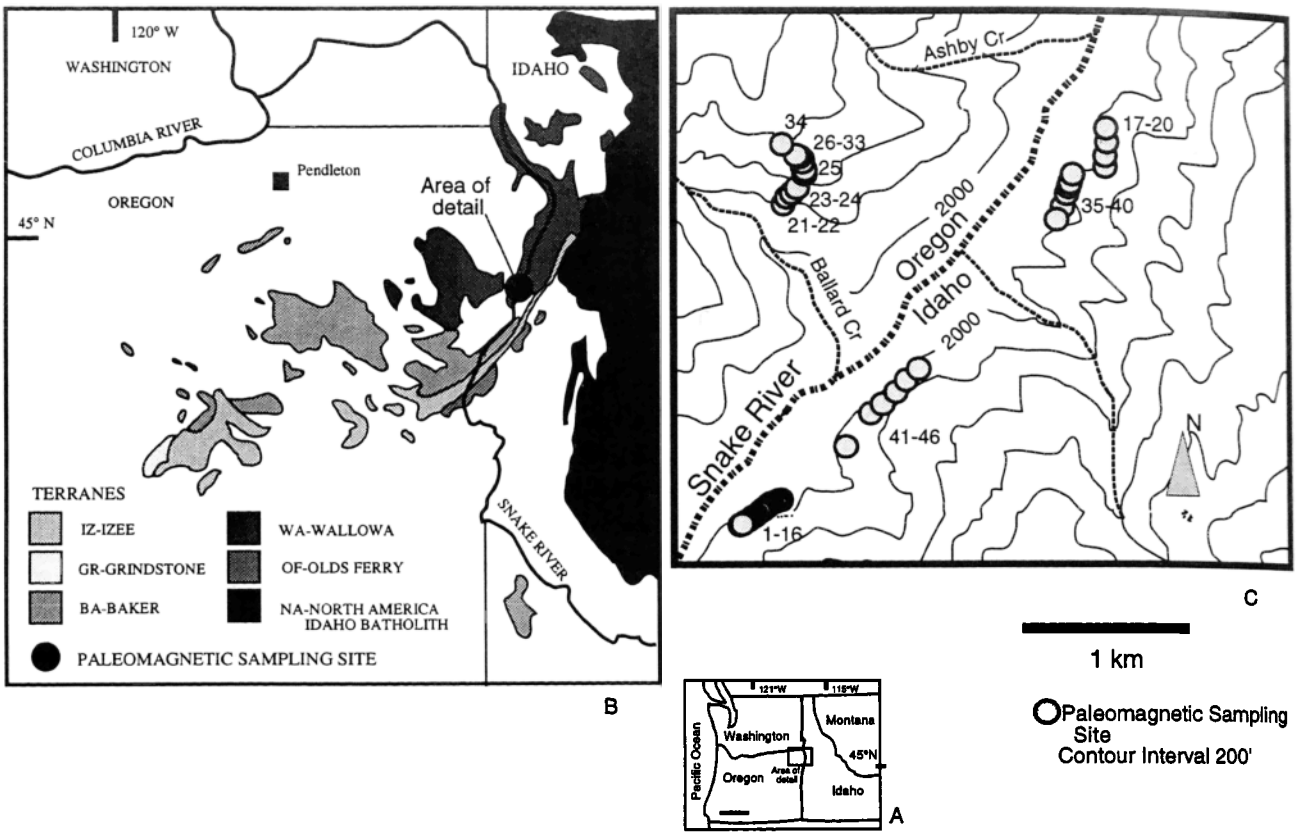


Figure 1. Geological sketch map of the Hells Canyon region of eastern Oregon. (a) Regional geography, (b) simplified terrane map of this region and, (c) topographic map showing the location of our sampling sites. Please refer to Table 1 for a more detailed description of individual paleomagnetic sites. From 3 to 10 paleomagnetic cores were collected at each sampling site.

al., 1982]. If the lower part of the Seven Devils Group was deposited in the northern hemisphere, we would then expect predominantly upward inclinations from paleomagnetism; successful measurement of the Permian magnetization would constrain displacement estimates for the Seven Devils Group.

Jones et al. [1977] interpreted the Seven Devils Group as part of the Wrangellia terrane, a distinctive sequence of Carboniferous and Permian volcanic arc rocks overlain by a thick section of Middle and Upper Triassic basalt. The Wrangellia terrane is broken into a number of large blocks located in British Columbia, in southeastern Alaska, and in the Alaska Range. Paleomagnetism of the Triassic basalts of the Wrangellia terrane indicated that the Alaskan blocks had moved northward relative to cratonic North America more than 28° of latitude, probably from the equatorial zone near 15°N [*Hillhouse and Gromme*, 1984]. The paleolatitudes for the Triassic parts of the Wrangellia terrane on Vancouver Island [*Yole and Irving*, 1980] and in Hells Canyon are similar to paleolatitudes of the Alaskan fragments, although the inferred displacements are much smaller for the two southern blocks. If the Wrangellia terrane extends into Hells Canyon, as proposed by *Jones et al.* [1977], then by solving the hemisphere ambiguity, paleomagnetic results from the Permian strata of the Seven Devils Group would set limits on displacement estimates for a belt extending from Oregon to the Alaska Range. Considering the central Alaskan part of the Wrangellia terrane, the difference of paleolatitude between the northern and southern hemisphere options is 30°, or about 3300 km of displacement.

Opinion has varied, however, on the validity of interpretations that consider the Seven Devils Group to be part of the Wrangellia terrane. Although the stratigraphic sections of the Seven Devils terrane and Wrangellia terrane of Alaska and British Columbia are similar in terms of age, paleontology, and paleomagnetism, clear differences have been noted in volcanic chemistry between the Seven Devils Group and the Wrangellia terrane proper. For example, *Sarewitz* [1983] pointed out that the calc-alkalic andesites and tuffs of the Seven Devils Group contrast sharply with the dominantly tholeiitic basalts of the Wrangellia terrane on Vancouver Island and in Alaska. Although the chemistry implies different tectonic settings for the volcanic outpourings, the age progression of strata, particularly in the Late Triassic, implies a close spatial association. *Jones et al.* [1977] noted the strong similarity of fossil molluscs in the Triassic limestones, which cap the thick volcanic sequences throughout the Wrangellia terrane. In particular, the transition from volcanism to carbonate deposition is coeval according to the paleontology of the Seven Devils Group and the Wrangellia terrane proper. In view of the uncertain correlation with the Wrangellia terrane, the Seven Devils Group has been designated part of the Wallowa terrane by *Silberling and Jones* [1984]. If not identical, the two terranes were probably close together during the Late Triassic.

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Geologic Framework of the Wallowa Terrane

Pre-Tertiary rocks in the Blue Mountains region of eastern Oregon and western Idaho formed in a volcanic arc [*Hamilton*, 1963]. This arc, named the Blue Mountains island arc by *Vallier and Brooks* [1986], consists of five major tectonostratigraphic terranes [*Brooks and Vallier*, 1978; *Dickinson*, 1979; *Vallier*, 1995] which were named the Baker, Grindstone, Izee, Olds Ferry, and Wallowa terranes by *Silberling and Jones* [1984]. The terranes were placed in their original plate tectonic settings by *Vallier* [1995], who concluded that the Baker and Grindstone terranes are parts of a complex forearc region of an intraoceanic island arc, the Wallowa and Olds Ferry terranes are parts of the volcanic or magmatic axes, and the Izee terrane is an intra-arc basin that in part overlapped the Baker, Grindstone, and Olds Ferry terranes. The Wallowa terrane was an active volcanic axis in the Early Permian and during the Middle and Late Triassic (Ladinian and early Karnian) interval. According to *Vallier* [1995], the volcanic axis of the arc changed in the late Karnian and a new axis (Olds Ferry terrane) developed after a probable change in plate convergence. Furthermore, the terranes were amalgamated by the Late Jurassic and most of the accretion to North America occurred in the latest Jurassic and Early Cretaceous interval [*Vallier*, 1995]. The accretion of the arc to North America caused metamorphism only near the suture [*Criss and Fleck*, 1987].

Paleontological investigations of the Martin Bridge Limestone and the overlying Hurwal and Coon Hollow Formations, the latter of Callovian and Oxfordian age, have yielded important constraints on the paleogeography of the Wrangellia terrane [*Newton*, 1986; *Stanley*, 1986; *Vallier and Brooks*, 1986]. *Senowbari-Daryan and Stanley* [1988] have shown that the Tethyan sponges *Amblysiphonella* cf. A. *steinmanni*, and *Colospongia whaleni* n. sp. occur within the Martin Bridge Limestone. The Wallowa terrane is the only site in North America where these fauna are known, strongly supporting the allochthonous terrane model. In addition, *Stanley and Whalen* [1989] have shown faunal similarities among sponges of Late Triassic age between the Wallowa terrane and units in the Koryak composite terrane (common species are *Retiophyllia dawsoni, Kuhnastraea cowichanensis, R. vesiculosa, and Storesia libratosepta*) of northeastern Russia linking allochthonous terranes of the eastern and western Pacific basin [*Stanley and Whalen*, 1989]. Investigations of the fossils in the overlying Jurassic units have shown similarities in coral and bivalve fauna between the Wallowa terrane and the Seven Devils arc terrane and the Western Interior Embayment of the North America plate, strongly implying a close proximity of these units by the end of the Middle Jurassic [*Stanley and Beauvais*, 1990].

In this paper we focus attention on the Wallowa terrane, particularly that part exposed in Hells Canyon of the Snake River. The terrane has a rather simple gross stratigraphy that changes irregularly across the region (Figure 2). Pre-Tertiary strata are mostly Permian and Triassic volcanic flow and volcaniclastic rocks overlain unconformably by Late Triassic platform limestone and Triassic to Jurassic siliciclastic basinal sediments. The Seven Devils Group of *Vallier* [1977] consists of the Early Permian Windy Ridge and Hunsaker Creek formations and the Middle and Late Triassic (Ladinian and early Karnian) Wild Sheep Creek and Doyle Creek Formations. The Triassic rocks in the Seven Devils Group were previously sampled for paleomagnetic studies [*Hillhouse et al.*, 1982] and the Permian Hunsaker Creek rocks were sampled for this report. Plutonic rocks both underlie and intrude the Pre-Tertiary sequence [*Vallier*, 1995]. Oldest plutonic rocks are metamorphosed dikes and small plutons of Late Pennsylvanian and Early Permian ages, some of which are crystallized magma chambers of overlying Permian volcanic strata. Youngest plutons are of Late Jurassic and Cretaceous ages, the largest of which is the Wallowa batholith. No large plutons cut the strata in the Hells Canyon region.

Geologic Setting of the Hunsaker Creek Formation, Hells Canyon

The Permian strata in the Hunsaker Creek Formation of Hells Canyon generally strike northeast and dip northwest. Folds are open and broad, and faults are confined to narrow shear zones. The rocks were deposited and erupted in a shallow marine environment (probably shelf and upper slope) and contain calcareous fossils (mostly brachiopods, the genus *Megousia* being dominant) which indicate an Early Permian age. The rocks are mostly volcaniclastic. In places, tuffaceous rocks predominant, whereas in others the sequences are conglomerate, sandstone, and siltstone. The thickness of strata varies within the terrane, both because of the original changes of lithofacies and the subsequent irregular uplift and erosion. Maximum thickness, however, is estimated to be about 3000 m [*Vallier*, 1977].

Greenschist metamorphism altered the predominantly siliceous and intermediate subalkaline rocks of the Hunsaker Creek Formation [*Vallier and Batiza*, 1978; *Vallier*, 1995]. Original quartz, plagioclase feldspar (oligoclase to labradorite), pyroxene (ortho- and clinopyroxene), glass, iron ore minerals, and minor amphibole were altered to an assemblage of albite, quartz, chlorite, calcite, epidote ±sphene, ±white mica, ±actinolite (mafic rocks), ±prehnite, ±pyrite. Pressures and temperatures were relatively low, probably in the 1-2 kbar and 250°-350°C range, respectively. The Permian rocks have a more equilibrium greenschist mineral assemblage than the overlying Triassic strata of the Seven Devils Group [*Vallier and Batiza*, 1978; *Vallier*, 1995] which are in the lower greenschist (dominant) and zeolite facies. Metamorphism is related both to burial and to the later crystallization of the Jurassic to Cretaceous plutons.

Accretion of the Wallowa terrane to the North America plate probably occurred as early as the Late Jurassic to the Early Cretaceous (recorded by deformation of the John Day inlier [*Dickinson*, 1979]). However, accretion might have occurred as late as 135 to 80 Ma on the basis of $^{40}Ar/^{39}Ar$ ages from granitic rocks within and near the suture zone between the Wallowa terrane and the North America plate [*Criss and Fleck*, 1987], and development of mylonitic foliation along the western Idaho suture zone [*Strayer et al.*, 1989].

Paleomagnetism of the Hunsaker Creek Formation

Samples were collected from well-exposed turbidite beds and rare sills of the Hunsaker Creek Formation (approximate site location, latitude 45.04°N, longitude 243.22°E) along the

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Table 1. Site Locations

Site Name	Site Cores	Strike, Dip	Lithology	Number of Line Segments by Temperature		
				Low	Medium	High
86HEL01	6Z001-005	52, 45S	sedimentary	2	5	1
86HEL02	6Z006-008	218, 60N	sedimentary	0	1	0
86HEL04	6Z009	236, 62N	sedimentary	0	0	0
86HEL05	6Z010-011	228, 58N	sedimentary	1	2	0
86HEL06	6Z012-014	228, 58N	sedimentary	2	2	1
86HEL07	6Z015-017	230, 57N	sedimentary	1	2	0
86HEL08	6Z018-021	232, 58N	sedimentary	3	4	0
86HEL09	6Z022-025	227, 61N	sedimentary	3	5	0
86HEL10	6Z026-033	230, 66N	sedimentary	6	8	0
86HEL11	6Z034-037	230, 66N	sill	5	8	0
86HEL12	6Z038-042	230, 48N	sedimentary	4	5	0
86HEL13	6Z043-047	230, 57N	sedimentary	2	2	1
86HEL14	6Z048-053	224, 58N	sedimentary	4	4	5
86HEL15	6Z055-058	224, 58N	sedimentary	3	4	2
86HEL16	6Z059-064	232, 66N	sedimentary	3	6	1
86HEL17	6Z065-069	265, 8N	sedimentary	4	6	0
86HEL18	6Z070-074	265, 8N	sedimentary	4	4	1
86HEL19	6Z075-081	237, 13N	sedimentary	0	1	1
86HEL20	6Z082-084	232, 12N	sedimentary	1	3	2
86HEL21	6Z085-090	270, 27N	sedimentary	1	6	2
86HEL22	6Z091-096	270, 27N	dike	1	5	0
86HEL23	6Z097-102	249, 27N	sedimentary	4	5	4
86HEL24	6Z103-107	249, 27N	sedimentary	2	5	5
86HEL25	6Z108-112	248, 29N	sedimentary	1	5	3
86HEL26	6Z113-117	252, 38N	sedimentary	2	5	0
86HEL27	6Z118-122	252, 38N	sedimentary	3	5	3
86HEL28	6Z123-126	252, 38N	sedimentary	2	2	2
86HEL29	6Z127-132	252, 38N	sedimentary	1	5	0
86HEL30	6Z133-135	252, 38N	sedimentary	3	3	1
86HEL31	6Z136-140	252, 38N	sedimentary	1	3	4
86HEL32	6Z141-146	252, 38N	sedimentary	0	7	0
86HEL33	6Z147-151	260, 37N	sedimentary	2	3	3
86HEL34	6Z152-155	272, 48N	sedimentary	2	3	0
86HEL35	6Z156-161	35, 8S	sedimentary	1	6	0
86HEL36	6Z162-166	35, 8S	sedimentary	2	2	2
86HEL37	6Z167-172	35, 8S	sedimentary	1	6	4
86HEL38	6Z173-178	20, 5S	sedimentary	3	3	2
86HEL39	6Z179-184	184, 11N	sedimentary	1	3	0
86HEL40	6Z185-189	20, 13N	sedimentary	2	4	1
86HEL41	6Z190-196	95, 61N	sedimentary	1	3	0
86HEL42	6Z197-201	207, 53N	sedimentary	0	0	0
86HEL43	6Z202-207	199, 57N	sedimentary	3	5	0
86HEL44	6Z208-217	199, 57N	sedimentary	3	7	1
86HEL45	6Z218-222	197, 51N	sedimentary	1	1	0
86HEL46	6Z223-227	172, 22N	sedimentary	1	1	0

Site location, site name, site cores, strike (0° to 360°), and dip (0° to 90°) used to transform between uncorrected and corrected coordinates, the lithology of the site and the number of line segments isolated during stepwise thermal demagnetization at low, medium and high temperatures. Corresponding directions and statistical parameters for these magnetizations are given in Table 2. Ninety-two samples show a linear, low-temperature magnetic component, 175 samples show a linear medium-temperature component and 52 samples show a linear magnetic component during the highest temperature thermal demagnetization steps.

Snake River in Oregon and Idaho (Figure 1 and Table 1). Between three to six oriented cores were collected from each selected horizon, defined as a sampling site. Azimuths for the core orientations were determined by magnetic and solar readings. Each core was cut into one or occasionally two paleomagnetic specimens.

We applied thermal and alternating field (AF) demagnetization methods to assess the components of

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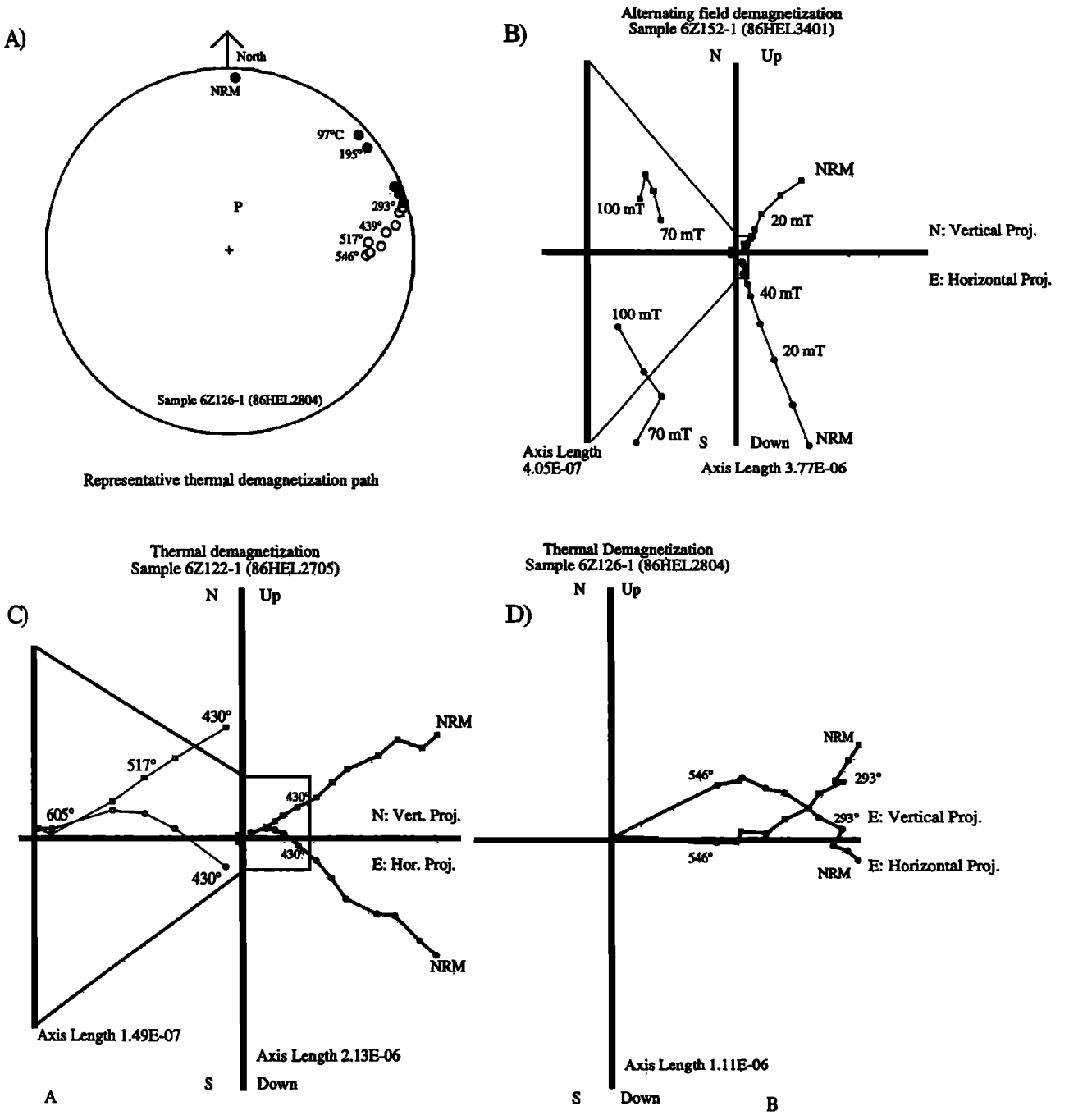


Figure 2. Some demagnetization characteristics of our samples. (a) Demagnetization planes were used to constrain the direction of magnetic component B but were too scattered to accurately constrain magnetic component C. (b) Orthogonal vector plot showing behavior of specimen 6Z152-1 during alternating field demagnetization to 100 mT (right) and expanded plot of 70-100 mT treatment (left). Note the single component behavior during stepwise AF demagnetization. In this and other orthogonal vector plots the horizontal projection is shown with open squares, while the vertical projection of the demagnetization data is shown with open circles. (c) and (d) Thermal demagnetization behavior for representative samples. Note the change from lower hemisphere downward directed magnetic vectors to upper hemisphere, negative inclination magnetic directions.

magnetization in a number of pilot specimens. Orthogonal projections of the demagnetization results were examined for vector components, which in turn were estimated by the principal component method [*Kirschvink*, 1980]. Alternating field treatment to 100 mT failed to isolate characteristic magnetizations (Figure 2) and thermal demagnetization proved to be the better treatment. A specimen from each core was demagnetized at 12-15 heating steps, which revealed two

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Table 2. Results of the Paleomagnetic Study

Site	n_A	I_A	D_A	k_A	n_B	I_B	D_B	k_B	n_C	I_C	D_C	k_C
86HEL01	2	72.4	105.2		5	84.0	281.9	7.5	1	-70.6	150.4	
86HEL02					1	67.1	32.5					
86HEL05	1	54.8	356.0		2	54.9	13.8					
86HEL06	2	22.5	95.0		2	24.9	75.5		1	-62.8	60.7	
86HEL07	1	-7.3	129.5		2	7.7	230.7					
86HEL08	3	84.0	44.9	13.2	4	69.4	268.7	8.4				
86HEL09	3	38.6	312.1	9.7	5	51.7	305.5	2.1				
86HEL10	6	75.5	71.1	6.6	8	51.5	294.9	5.9				
86HEL11	5	38.3	125.9	3.6	8	75.6	52.0	261.3				
86HEL12	4	75.1	31.9	13.0	5	70.9	26.9	25.3				
86HEL13	2	47.7	56.1		2	71.5	65.7		1	-35.7	79.0	
86HEL14	4	64.9	26.3	4.7	4	-45.8	169.5	8.4	5	-19.8	359.4	4.0
86HEL15	3	40.7	67.3	3.3	4	46.8	21.5	8.3	2	-17.8	333.1	
86HEL16	3	1.5	115.5	5.7	6	-13.9	71.0	2.2	1	-33.7	0.8	
86HEL17	4	60.6	40.5	4.4	6	69.0	12.2	6.3				
86HEL18	4	10.8	82.1	7.6	4	33.8	42.4	6.8	1	-24.1	59.0	
89HEL19					1	56.1	31.2		1	-23.9	43.4	
86HEL20	1	48.0	27.3		3	57.5	31.2	102.9	2	-28.8	47.6	
86HEL21	1	57.0	335.6		6	62.5	2.3	211.8	2	-46.8	2.0	
86HEL22	1	58.2	349.1		5	65.1	358.2	2707.7				
86HEL23	4	61.6	15.0	13.8	5	31.3	292.4	1.7	4	-41.9	60.7	253.7
86HEL24	2	59.7	293.6		5	-67.3	12.3	4.0	5	-46.7	62.9	395.7
86HEL25	1	64.6	263.3		5	64.9	359.1	126.1	3	-55.1	38.1	50.2
86HEL26	2	56.9	26.1		5	53.8	44.6	49.4				
86HEL27	3	38.2	59.6	10.5	5	47.2	48.6	116.9	3	-7.4	44.9	26.2
86HEL28	2	38.2	22.0		2	32.3	55.5		2	-20.1	85.0	
86HEL29	1	45.6	50.4		5	68.3	28.0	1221.6				
86HEL30	3	63.5	344.8	12.2	3	48.2	330.8	24.5	1	-12.2	354.8	
86HEL31	1	56.3	35.1		3	51.8	31.2	245.1	4	-27.2	11.9	29.1
86HEL32					7	64.5	15.3	132.0				
86HEL33	2	-34.6	86.6		3	-40.5	107.3	4.2	3	-27.2	96.0	10.8
86HEL34	2	83.8	349.8		3	71.3	82.5	1.8				
86HEL35	1	10.6	92.2		6	72.7	1.1	2.3				
86HEL36	2	38.9	78.7		2	17.7	111.5		2	-39.5	113.0	
86HEL37	1	83.8	120.4		6	87.4	289.5	8.4	4	-48.6	76.5	17.9
86HEL38	3	19.5	17.4	7.1	3	7.7	38.6	2.3	2	-37.1	52.8	
86HEL39	1	48.9	320.8		3	27.8	342.7	2.2				
86HEL40	2	46.2	57.2		4	-29.6	40.6	8.6	1	-32.7	57.7	
86HEL41	1	-28.5	93.4		3	54.5	51.4	2.9	1	1.7	170.9	
86HEL43	3	78.4	112.6	5.5	5	22.5	297.5	35.0				
86HEL44	3	40.8	289.3	3.3	7	29.7	271.9	3.0				
86HEL45	1	43.7	94.4		1	29.4	19.7					
86HEL46	1	14.0	65.0		1	55.8	357.9					

A, low, B, medium, and C, high temperature remanent magnetic components. Site name, n is number of samples which displayed a particular magnetic component, I and D, uncorrected inclination and declination, and k, is the Fisher estimate of kappa.

and sometimes three components of magnetization. Results of the principal component analysis are shown in Table 2.

Alternating field demagnetization showed a low medium destructive field or 10 to 40 mT (Figure 3). Isothermal remanent magnetization (IRM) experiments showed rapid saturation at low applied fields (generally >20 mT) implying the presence of titanomagnetite as the dominant magnetic mineral in these samples.

Anisotropy of magnetic susceptibility (AMS) measurements of 93 selected samples shows incoherent grouping of AMS directions. Samples were poorly grouped when plotted as AMS eccentricity ellipsoid or AMS foliation versus lineation (Figure 4). Directions of K_{max}, K_{int} and K_{min} (the maximum, intermediate, and minimum axis of magnetic susceptibility) were also poorly grouped, even for the subset of samples in which the eccentricity ellipsoid of AMS was less than 1 (50 of 93 samples). The results were so randomly distributed in both corrected and uncorrected coordinates that the reviewers of this manuscript suggested deleting sterographic projections of these data.

The reviewers also suggested that we calculate statistics for the magnetic components using only those site which had n (samples) ≥ 3 and k (Fisher estimate of kappa) ≥ 10. We calculated mean directions for each magnetic component and unfolding statistics in three ways. In method 1 we include only sites which have $n \geq 3$ and an estimate of kappa ≥ 10. In method 2 we include sites which have $n \geq 3$, and method 3 uses all sites. For the A and B magnetic components each method gave roughly the same mean direction; however, the differences in unfolding statistics were unexpected for the C component.

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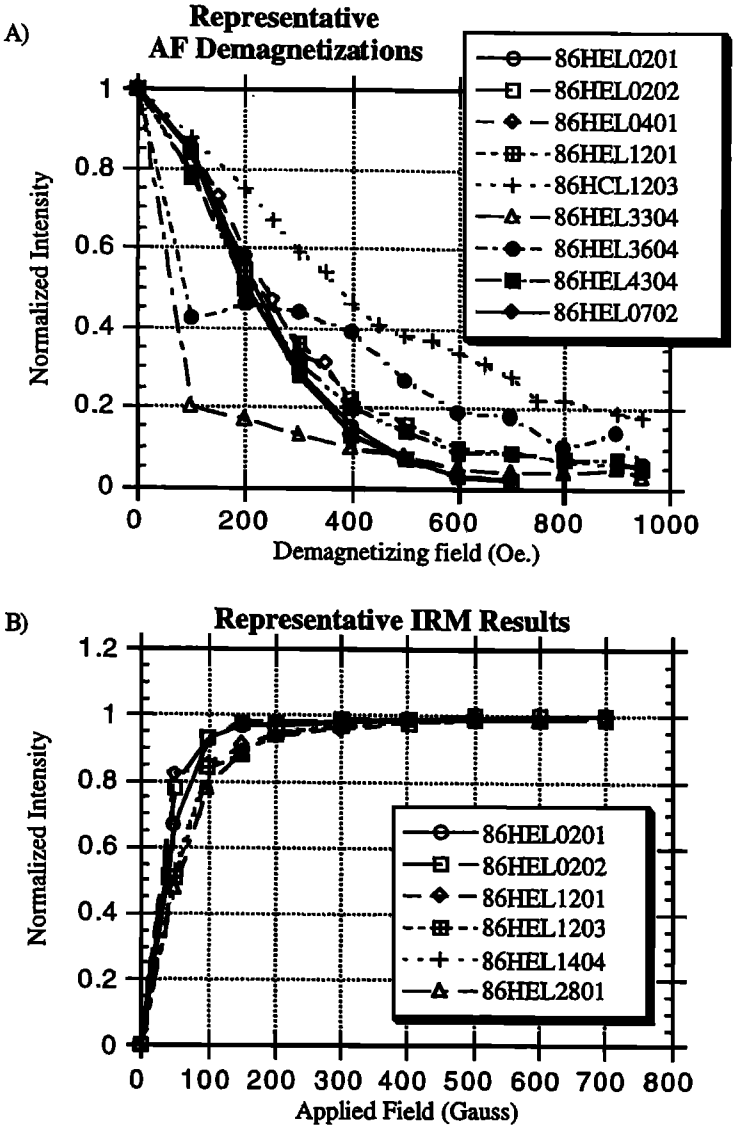


Figure 3. Representative variation in remanent intensity during (a) alternating field (AF) demagnetization and (b) isothermal remanent magnetization (IRM) experiments for selected samples. Note the relatively low median destructive field in Figure 3a, and the rapid saturation with applied field in Figure 3b, suggestive of magnetite as the dominant magnetic mineral present in these samples.

The lowest temperature component (A) was observed in 92 specimens from 40 sites by heating below 197°C. The site mean directions for component A are strongly streaked. Using method 1, described above, the mean direction is I=67.4°, D=29.5°, α_{95}=21.1°, k value=14.1, N=5. Correcting for tilt of the bedding causes poorer grouping of magnetic component A (Figures 5 and 6a and Tables 2 and 3).

Magnetic component B is observed during AF and thermal demagnetization (Table 3). Among 43 sites, 175 specimens displayed this magnetic component, which unblocks during heating steps of between ≈200°C and ≈550°C and is generally removed by 580°C (Figure 5a). The Fisherian mean for component B, using method 1, in uncorrected coordinates is I=64.8°, D=10.3°, k value=10.6, α_{95}=13.4°, N=13 (Figure 6b). Bingham statistics were used to calculate the location of this direction from the poles to planes fit to the demagnetization data when the magnetic directions followed great circles during thermal demagnetization [*Kirschvink*, 1980]. The Bingham estimate calculated from linear magnetic components before correcting for the tilt of the beds agreed with the mean direction calculated from the intersection of in situ demagnetization planes (which is I=70.6°, D=13.0°, n(specimens)=102, using method of *Kirschvink* [1980]). Sites which display this magnetic component become increasingly scattered during unfolding (Figures 5 and 6b).

The highest unblocking temperature magnetic component, magnetic component C, was observed in a relatively small number of samples (Table 2). The C magnetic component is observed to unblock between thermal demagnetization temperatures of ≈500°C and ≈680°C (Figure 5a). At 22 sites, a total of 51 specimens, or about 30% of the specimens that displayed a coherent magnetic direction between heating steps, displayed a C magnetic component during stepwise thermal demagnetization. This magnetic component is always of reversed polarity in both in situ and tilt-corrected coordinate systems.

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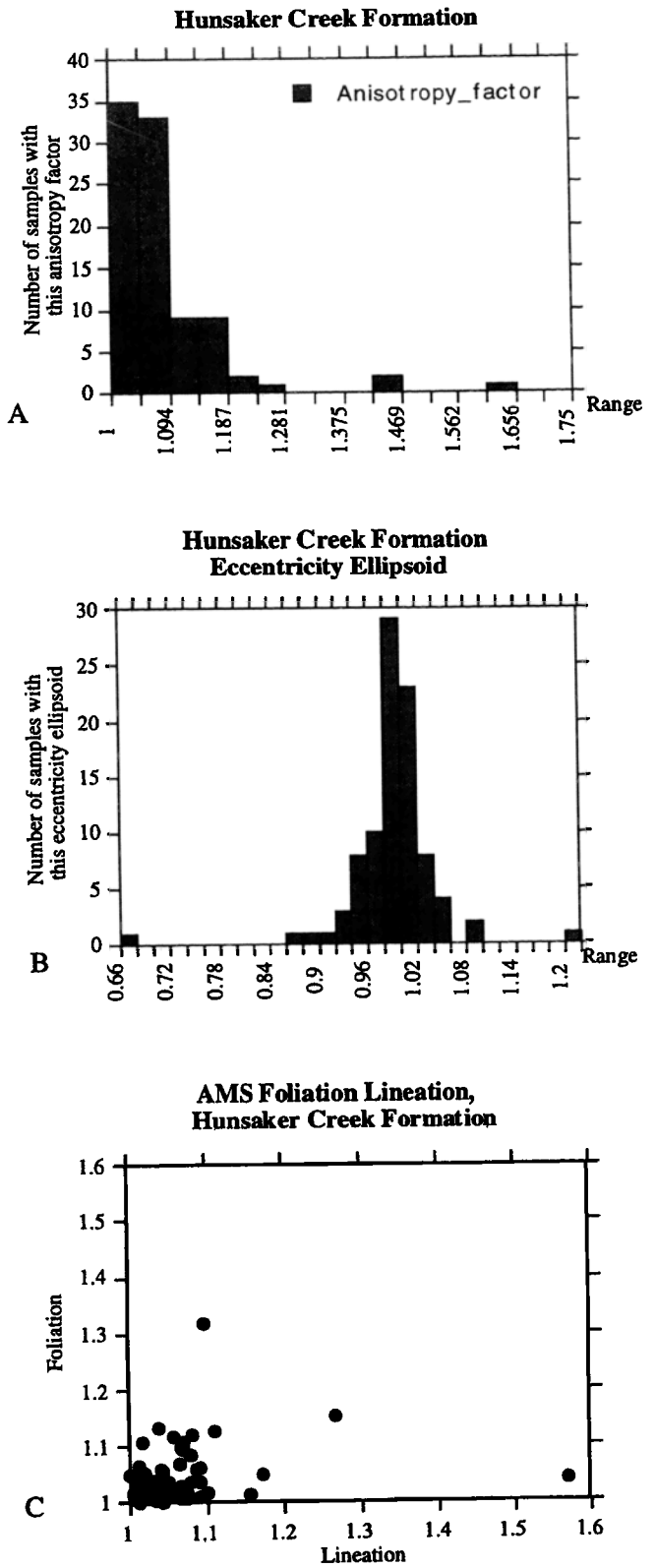


Figure 4. Anisotropy of magnetic susceptibility related data for the Hunsaker Creek Formation. (a) The anisotropy factor (K_{max}/K_{min}) for subset of 93 samples for which we measured AMS. (b) The eccentricity ellipsoid $(K_{int})^2/(K_{max}*K_{min})$. Note that our samples show only weak anisotropy and both oblate and prolate. (c) AMS lineation (K_{max}/K_{int}) is plotted with respect to variation in AMS foliation (K_{int}/K_{min}).

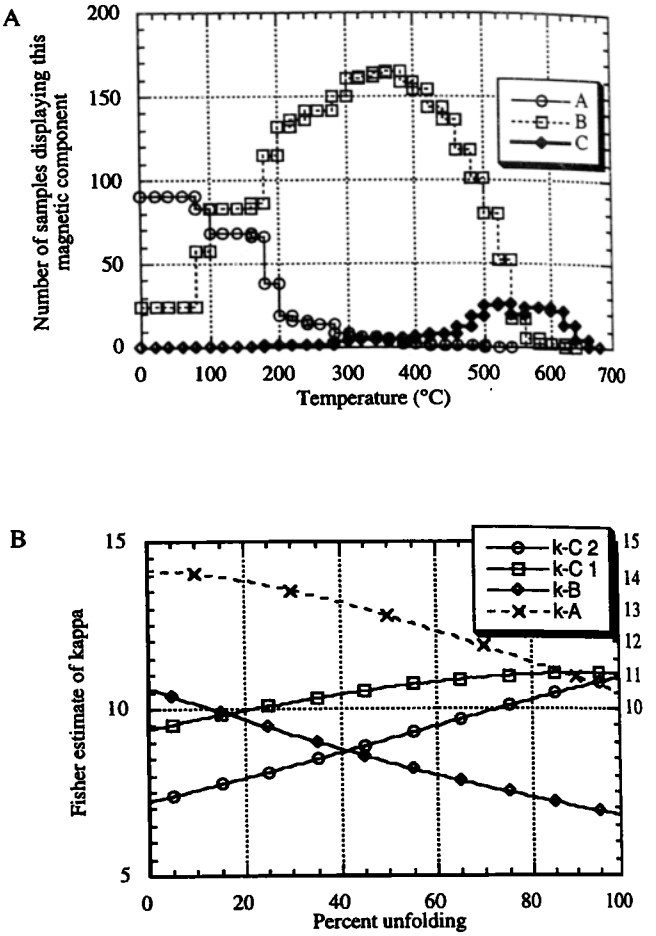


Figure 5. (a) Unblocking spectra for magnetic components A, B, and C. Principal component analysis was used to determine best fitting lines to the stepwise thermal demagnetization data. The initial and ending demagnetization step temperatures for each linear component were then used to construct this histogram. Note that magnetic component B is the best constrained, while in many fewer specimens only at higher temperature was magnetic component C observed. (b) Incremental fold tests on magnetic components A, B, and C 1, using method 1 (only sites with $n \geq 3$, $k \geq 10$ used in calculating Fisher mean and associated statistics), and C 2 (only sites with $n \geq 3$ used in calculating Fisher mean and associated statistics). Both magnetic components A and B appear to postdate folding. The best grouping of component C using method 1 or 2 occurs at 95% and 100% unfolding.

Using method 1, the overall mean for component C, as observed in seven sites, shows best grouping at 95% unfolding, although it is difficult to assess whether this small increase in the overall k value is meaningful. Using method 2, best grouping is observed for eight sites at 100% unfolding. While using method 3, for 22 sites, best grouping is observed at 67% unfolding. Using method 1, best grouping is observed at an overall 95% unfolding, the corresponding overall Fisherian mean at 95% of unfolding is I= -45.2°, D= 77.3°, k value=11.1, and α_{95}= 19.0° (Figures 5, 6c, and 6d).

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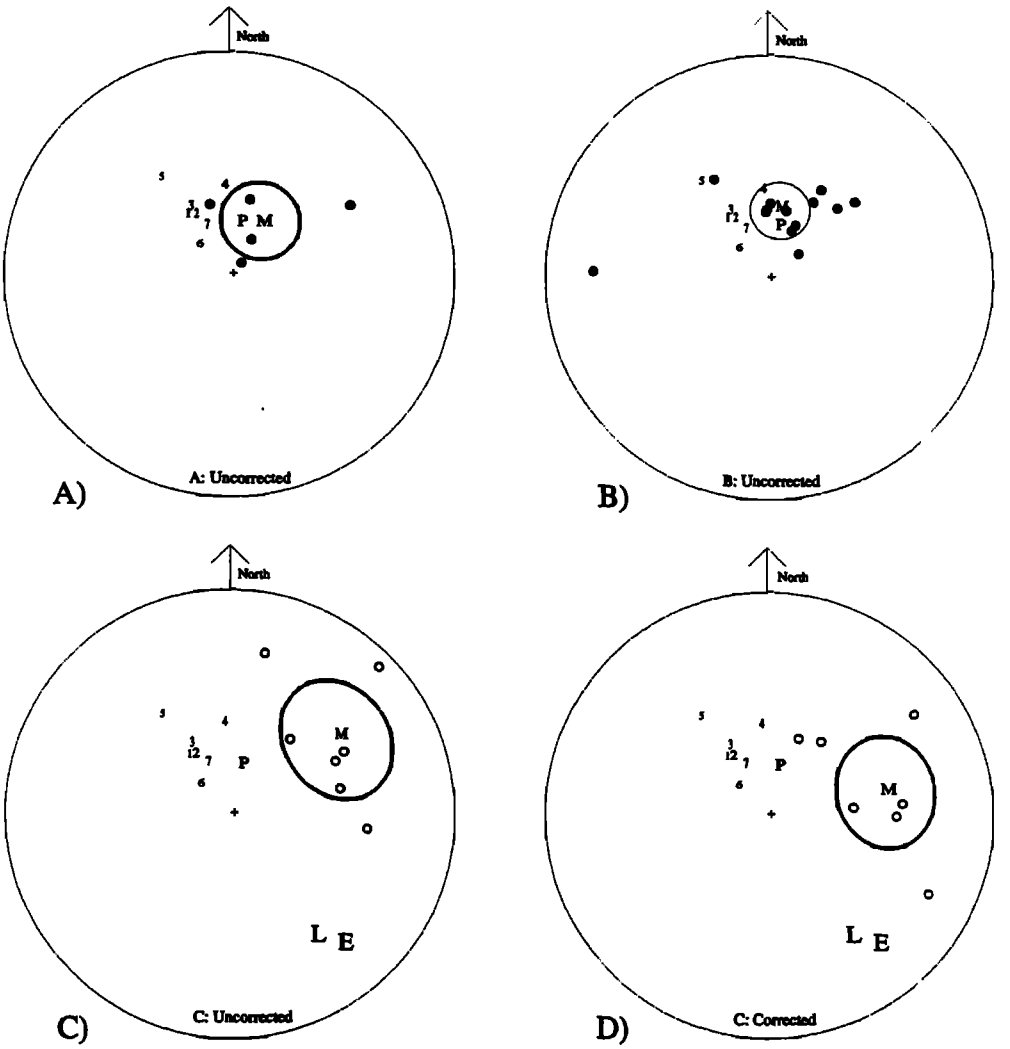


Figure 6. (a) Equal-area stereographic projection showing the mean direction (M), and associated Fisher 95% circle of confidence for the secondary A characteristic magnetization before unfolding compared with the expected present-day magnetic direction "P" and the expected Cretaceous magnetic directions. Expected directions are 1, *Harrison and Lindt* [1982] 135 Ma APWP pole; 2, *Irving and Irving* [1982] 131 Ma APWP pole; 3, *Bazard and Butler* [1994]; 4, *Van Fossen and Kent* [1990]; 5, *Bazard and Butler* [1992]; 6, *Van Fossen and Kent* [1993]; and 7, *Foster and Symons* [1979]. Open circles are upper hemisphere directions; solid symbols are lower hemisphere magnetic directions. (b) Equal-area stereographic projection showing the secondary B characteristic magnetization before unfolding compared with the expected present-day magnetic direction and the expected directions calculated using the reference poles and symbols of Figure 6a. (c) Equal-area stereographic projection showing the C characteristic magnetization in geographic coordinates compared with present-day field, and expected Cretaceous/Late Jurassic directions used in Figures 6a and 6b, and the upper hemisphere expected directions calculated using the average reference poles for E, Early Permian, and L, Late Permian. Open circles are upper hemisphere (reverse polarity) directions; the mean direction, "M", is shown surrounded by the Fisher 95% circle of confidence. (d) Equal-area stereographic projection showing the mean C characteristic magnetization in stratigraphic coordinates, represented by the "M", is compared with present-day field, expected Cretaceous/Late Jurassic directions used in Figures 6a and 6b, and the upper hemisphere expected directions calculated using the average reference poles for E, Early Permian, and L, Late Permian.

Reference Paleomagnetic Studies of Rocks of Cretaceous and Early to Late Permian Age From the North America Craton

Highly reliable paleomagnetic reference poles from the stable North America craton are required to interpret the tectonic implications of the paleomagnetic poles obtained in our study. We used the 120-150 Ma poles from the apparent polar wander paths (APWP) for the North America plate of *Irving and Irving* [1982] and *Harrison and Lindh* [1982] and individual paleomagnetic poles from *Foster and Symons* [1979], *Van Fossen and Kent* [1990], *Bazard and Butler* [1992], *Van Fossen and Kent* [1993], and *Bazard and Butler* [1994] to

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Table 3. Mean Results for the A, B, and C Characteristic Remanent Magnetic Components Calculated in This Study

Component	*I*	*D*	*k*	α_{95}	*N*
A-(U)-1	67.4	29.5	14.1	21.1	5
A-(U)-2	64.4	48.6	4.2	20.7	16
A-(U)-3	65.2	44.7	4.0	13.1	40
A-(C)-1	33.4	352.5	10.5	24.7	5
A-(C)-2	37.3	353.1	3.6	23.0	16
A-(C)-3	-26.4	154.9	2.4	18.9	40
B-(U)-1	64.8	10.3	10.6	13.4	13
B-(U)-2	67.7	345.0	5.2	12.2	33
B-(U)-3	64.0	14.3	4.5	11.6	43
B-(C)-1	34.0	3.4	9.6	14.1	13
B-(C)-2	-25.5	166.8	3.1	17.1	33
B-(C)-3	-26.4	169.7	3.3	14.2	43
C-(U)-1	-41.1	54.7	9.4	20.8	7
C-(U)-2	-40.0	46.4	7.2	22.1	8
C-(U)-3	-41.5	45.7	5.7	14.2	22
C-(C)-1	-45.1	78.6	11.1	19.0	7
C-(C)-2	-46.2	73.7	10.9	17.5	8
C-(C)-3	-41.1	70.8	5.0	15.4	22
C-(O)-1	-45.2	77.3	11.1	19.0	7
C-(O)-2	-46.2	73.7	10.9	17.5	8
C-(O)-3	-46.3	59.8	6.9	12.7	22

The data are given as component, U (uncorrected), C (corrected), or O (optimum), 1, 2, or 3. Method 1 uses only sites given in Table 2 which have n(number of samples) ≥3 and k (Fisher estimate of kappa) ≥ 10. Method 2 uses only sites which have $n \geq 3$, and method 3 uses all the sites given for that specific magnetic component given in Table 2 regardless of quality. Other data are, I, Inclination, D, Declination, k, Fisher estimate of kappa, and α_{95} , Fisher alpha 95 confidence angle; N, number of sites used in calculation. For component A, using methods 1 and 2, the maximum estimate of kappa was observed at 0% unfolding. For method 3, the maximum estimate of kappa was observed at 4% unfolding. For component B the maximum estimate of kappa using grouping methods 1 or 2 was at 0% unfolding. For component C, using method 1, best grouping was observed at 95% unfolding, and using method 2, best grouping was observed at 100% unfolding.

calculate expected directions to compare with those observed. In the APWP data, paleomagnetic data collected from the stable North America craton are averaged within a moving window to yield a reference pole at 10-m.y. intervals.

Reference poles of Permian age from North America have been extensively reviewed and summarized [*Gordon et al.*, 1984; *Van der Voo,* 1990, 1993; *Pan and Symons,* 1993]. In their investigation of probable clockwise rotation, of the order of 3° to 9°, of the Colorado Plateau with respect to North America, *Pan and Symons* [1993] divided reference poles into groups corresponding to those collected from the interior craton (IC), Appalachian orogen (AO), and Colorado Plateau (CP) regions of North America.

In each of the studies, stepwise thermal demagnetization was used to isolate a high-temperature unblocking magnetic component. Because many of the rocks sampled for reference poles were formed during the Permian to Carboniferous reversed magnetic superchron, often the paleomagnetic reversal test could not be used to constrain their reliability. Furthermore, most of the sampled units are nearly horizontal making application of the fold test impossible. However, thermal demagnetization, remanent magnetization stability, the consistency of the polarity observed in these studies with deposition of sampled units during the reversed superchron, and the consistency of the reference studies with each other strongly suggest that these results are reliable reference apparent polar wander path points.

Reference poles given by *Pan and Symons* [1993] for Late Permian include one Appalachian orogen pole, one Colorado Plateau pole, and three interior craton poles. For Early Permian, three Appalachian orogen poles, seven Colorado Plateau, and four interior craton paleomagnetic poles are available, giving a total of 19 potential reference poles for the North America plate. Of these, the Colorado Plateau poles

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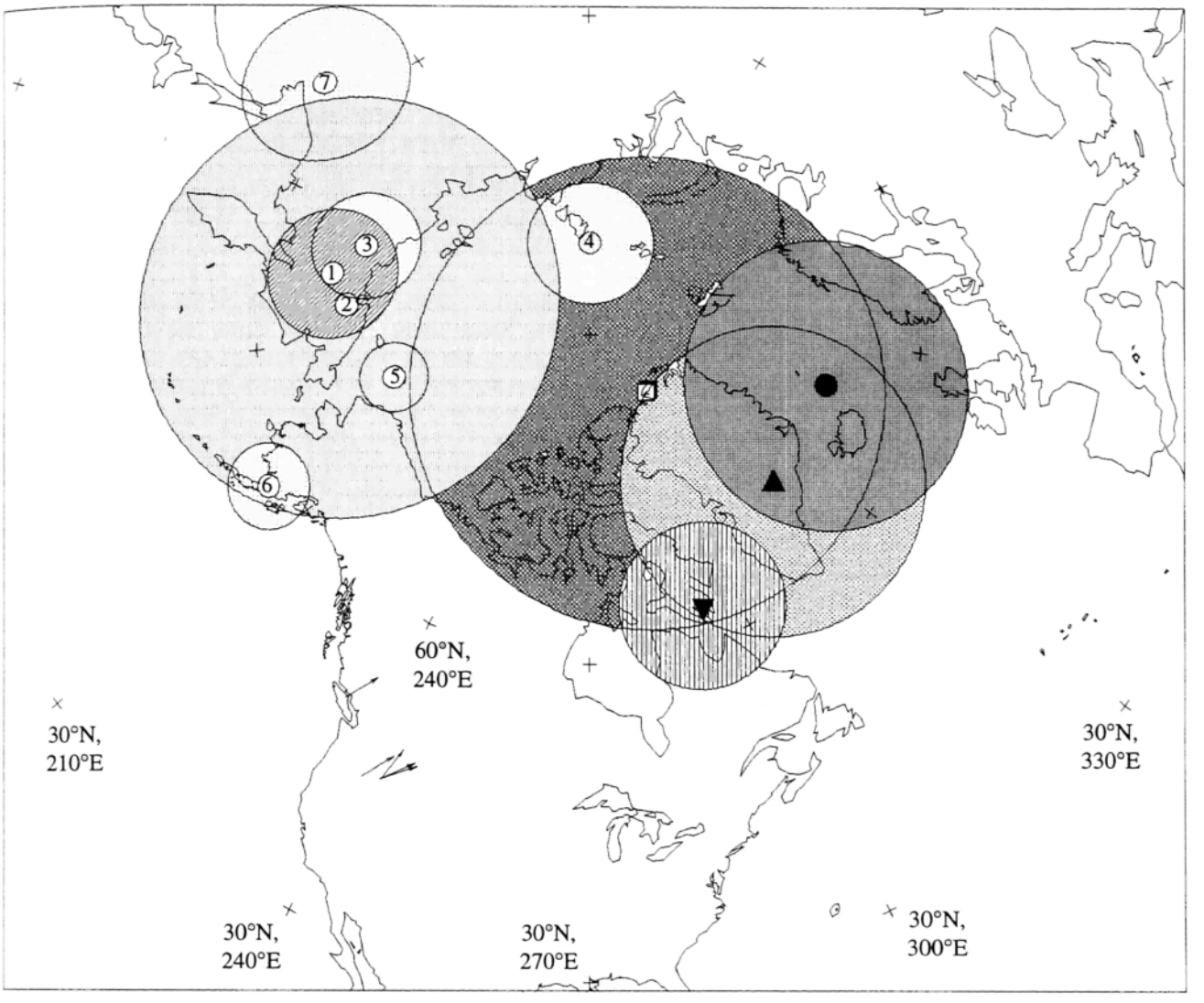


Figure 7. The paleomagnetic pole corresponding to magnetic component B (pole position calculated using method 1 is shown by open square with 95% confidence region) is compared with the expected reference poles 1, *Harrison and Lindt* [1982] 135 Ma APWP pole; 2, *Irving and Irving* [1982] 131 Ma APWP pole; 3, *Bazard and Butler* [1994]; 4, *Van Fossen and Kent* [1990]; 5, *Bazard and Butler* [1992]; 6, *Van Fossen and Kent* [1993]; and 7, *Foster and Symons* [1979]. In addition, the group 3 magnetizations of *Hillhouse et al.* [1982] (triangle) and Y magnetizations of *Yole and Irving* [1980] (solid circle), both of which failed fold tests, are shown. Each of these represents a postfolding secondary magnetization. These directions are compared with the paleomagnetic pole taken from *Wilson and Cox* [1980] of the Blue Mountains plutons of eastern Oregon of Late Jurassic to Early Cretaceous age (inverted triangle).

must be rotated counterclockwise to reconstruct them properly [*Steiner*, 1986; *Bryan and Gordon,* 1990]. In our analysis we rotated the Colorado Plateau paleomagnetic poles about the suggested Euler pole describing the relative rotation of the Colorado Plateau with respect to North America, of *Byran and Gordon* [1990] ($NAMR_{CPL}$ = 37°N, 103°W) and used the angle of rotation, applied to the paleomagnetic poles in a counterclockwise sense, given by *Pan and Symons* [1993] of 5.5°±2.9°. These rotated Colorado Plateau sites were then Fisher-averaged with the interior craton and Appalachian orogen sites, giving an average pole of λ=44.0°, ϕ=124.6°, A_{95}=2.7°, Fisher estimate of kappa=210.2, N=14, for the Early Permian. For the Late Permian, we rotated the single Colorado Plateau site about the pole given above and averaged with the paleomagnetic poles given in the summary of *Pan and Symons* [1993], to calculate an average paleomagnetic pole of λ=51.1°, ϕ=121.9°, A_{95}=4.9°, Fisher estimate of kappa=242.2, N=5.

A and B Characteristic Magnetization: Overprinting and Clockwise Rotation

Characteristic remanent magnetic components A and B appear to be of secondary origin as shown by their relatively lower, when compared with component C, unblocking temperatures, poorer grouping after unfolding, and similarity

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with the secondary group 3 (overprint) magnetization observed in the overlying Triassic rocks. The A (method 1 mean; I=67.4°, D=29.5°, α_{95}=21.1°, N=5) and B (method 1 mean; I=64.8°, D=10.3°, α_{95}=13.4°, N=13) magnetic components in geographic coordinates are similar to the present-day magnetic field direction for this site (PDF I=69.0°, D=17.7°). This similarity supports the interpretation that these magnetic components are being caused by overprints of the present-day magnetic field on the samples. However, an alternate interpretation is possible, noting the similarity of these overprinting magnetic directions and other paleomagnetic data collected.

If these magnetic components are interpreted as relating to reheating, these results are consistent with the 66° of clockwise rotation observed by *Wilson and Cox* [1980]. The paleomagnetic poles calculated from secondary magnetizations of the overlying Upper Triassic rocks and the Hunsaker Creek Formation are similar in location to the paleomagnetic pole calculated from the primary magnetization observed in the study of Late Jurassic to Early Cretaceous plutons in the Blue Mountains of eastern Oregon [*Wilson and Cox*, 1980] (Figure 7). The Blue Mountains plutons have tilt-corrected characteristic magnetization of I=63°, D=30° and α_{95}=10°. The paleomagnetic poles corresponding to magnetic components A and B are rotated significantly clockwise away from the APWP data for the North America plate. If we rotate the A and B poles back onto the APWP track, the data fall along the Cretaceous interval.

Yole and Irving [1980] sampled the Middle and Upper Triassic (late Ladinian to early late Karnian) Karmutsen volcanics of Vancouver Island. Basalt flows record two magnetizations, referred to as the X and Y magnetizations by them, within the sampled units. In 14 sites (66 oriented specimens) the normal-polarity Y magnetization appeared to represent a postfolding overprint, which showed significant clockwise rotation. As shown in Figure 7, paleomagnetic pole positions from Early Cretaceous rocks and overprints indicate clockwise rotation of the Blue Mountains, Vancouver Island, and Hells Canyon. We speculate that this consistent clockwise rotation could have been caused by deformation of this region in response to the significant component of right-lateral motion between the Kula-North America and Farallon-North America plates during the Late Cretaceous and early Tertiary [*Engebretson et al.*, 1984, 1985; *Debiche et al.*, 1987].

C Characteristic Remanent Magnetization: Northern Hemisphere Origin of Terrane and Possible Permian Left-Laterial Motion Along the Western North America Plate

The interpretation of magnetic component C is more complex than that of either A or B. Several hypotheses may be eliminated from examination of these data. Component C does not represent a secondary magnetization of Cretaceous age because such a secondary magnetic component is clearly represented by components A and B and removed. In addition, if inverted to upper hemisphere directions, these secondary characteristic magnetic components differ in declination with respect to component C by 180°. The hypothesis that component C represents a secondary Late Triassic to Cretaceous age magnetization can also be eliminated in that no such magnetization was observed in the immediately overlying Triassic rocks. *Hillhouse et al.* [1982] showed that the high unblocking temperature group 1 and 2 magnetizations, interpreted to have been acquired during deposition of the Middle and Upper Triassic Wild Sheep Creek Formation, passed a fold test (I=-31.1°, D=83.2°, α_{95}=17.9°, I=33.8°, D=318.6°, α_{95}=13.0°). The C magnetization (method 1; I=-45.2°, D=77.3°, α_{95}=19.0°, N=7) is similar to the group 1, reversed, magnetizations. An acceptable interpretation of component C data is that the magnetization was acquired during deposition of the overlying Wild Sheep Creek Formation during the Middle and Late Triassic, although using mean calculation methods 1 and 2, best grouping of our sites occurs at 95% and 100% unfolding. We reject the hypothesis that the C magnetization represents a secondary magnetic component acquired between Permian and Late Triassic on the basis of geological observations. Only a few outcrops of igneous rocks whose ages are bracketed between that of the Hunsaker Creek and Wild Sheep Creek Formations are known in this region [*Vallier*, 1995], and therefore significant thermal activity within that time interval is unlikely.

We prefer to interpret component C as being acquired during deposition of these units in the Early Permian because of (1) the removal of secondary Cretaceous A and B magnetizations, (2) the consistent magnetic polarity of the C magnetization seen in the sampled sections, and (3) the improved grouping of sites for which $n \geq 3$, $k \geq 10$, at 95%, and for sites with $n \geq 3$, at 100% unfolding. The consistency of

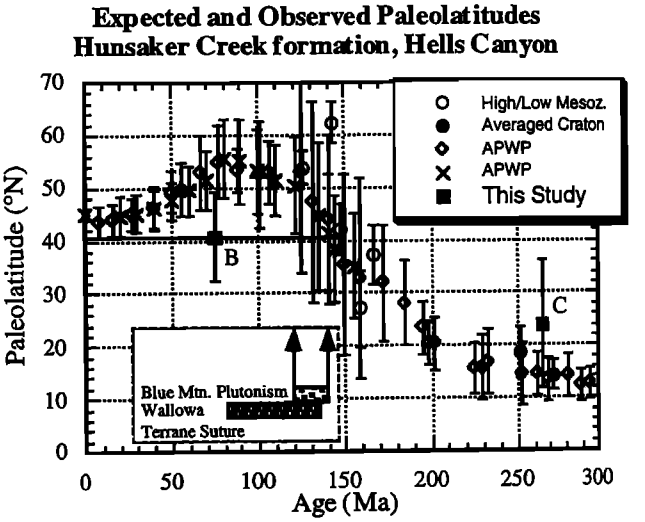


Figure 8. Comparison of paleolatitudes calculated using inclination-only statistics on our characteristic magnetic components B and C and associated uncertainties compared with expected paleolatitudes for this site calculated from North America plate reference studies. Symbols used are diamonds, *Irving and Irving* [1982]; crosses, *Harrison and Lindt* [1982]; open circles, individual paleolatitudes calculated from individual high and low latitude Mesozoic paleomagnetic poles; solid circles, averaged paleomagnetic poles for the Late and Early Permian; solid squares, our results. Tectonic events such as plutonism in the Blue Mountains and an estimate of the suture age of the studied terrane with the North America plate are also shown.

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Table 4. Reference Poles

Ref	Site	A_{min}	A_{max}	λ_p	ϕ_p	α_{95}
1	APWP	125		68	189	19
1	APWP	131		68	171	19
1	APWP	140		66	163	16
1	APWP	150		67	138	17
2	APWP	121		68.0	179.1	6.3
2	APWP	135		66.2	164.3	5.8
2	APWP	140		66.0	156.5	5.6
2	APWP	144		66.6	147.8	5.7
2	APWP	155		69.7	132.4	7.5
3	MON	126		72.0	191.0	3.1
4	NYK	143		58.0	203.1	3.8
5	BR	144	150	68.3	156.2	4.8
6	SU	158		56.3	133.4	7.2
7	MO	166		81.6	89.7	5.6
1	APWP	232		55	108	6
1	APWP	252		49	116	6
1	APWP	261		48	118	4
1	APWP	268		46	119	4
8	BS,PEI	Late	Perm	57*	128*	
8	Hus Tn	Late	Perm	50	121	
8	Och Rb	Late	Perm	55	119	
8	Gd Rb	Late	Perm	51	125	
8	DL Fm	Late	Perm	51	125	
	Avg.	Late	Perm	51.1	121.9	5.9
8	Up Pc	Early	Perm	42	127	
8	CM Fm	Early	Perm	45	129	
8	Mdl Pg	Early	Perm	35	132	
8	El Cn	Early	Perm	41*	126*	
8	Ct Fm	Early	Perm	41*	124*	
8	Hal Tn	Early	Perm	46*	124*	
8	Ct F C	Early	Perm	40*	128*	
8	Ct F U	Early	Perm	41*	122*	
8	FL Fm	Early	Perm	45*	126*	
8	MM F	Early	Perm	40*	121*	
8	Leo Ok	Early	Perm	52	119	
8	Cs Fm	Early	Perm	51	123	
8	Abo F	Early	Perm	50	118	
8	Ing F	Early	Perm	46	122	
	Avg.	Early	Perm	44.0	124.6	2.7

Ref refers to the source of the reference pole, site refers to the locality from which the reference pole was observed, A_{max} and A_{min} refer to the oldest and youngest age for which this pole is valid, and λ_p, ϕ_p, and α_{95} refer to the latitude, longitude, and cone of radius of 95% certainty about the reference pole. In reference 1 and reference 2, paleomagnetic data from stable North America craton was averaged within a moving time window. In addition, reference poles from individual studies of high reliability, references 3-8 (1, *Irving and Irving* [1982]; 2, *Harrison and Lindh* [1982]; 3, *Foster and Symons* [1979]; 4, *Van Fossen and Kent* [1993]; 5, *Bazard and Butler* [1994]; 6, *Bazard and Butler* [1992]; 7, *Van Fossen and Kent* [1990]; and 8, *Pan and Symons* [1993]) were used. Each of these studies has been accurately dated, appears to average secular variation, and comes from a stable region of the North America plate. For studies from the Colorado Plateau (λ_p*, ϕ_p*), the paleomagnetic pole given has been rotated using the Euler Pole location of *Byran and Gordon* [1990] and Euler rotation angle of *Pan and Symons* [1993], the rotated pole position is shown in this table. The Fisher estimate of kappa, for the Late Permian pole, is k=210.2 (N=5), and for the Early Permian k= 242.2 (N=14). We used these data to calculate the expected paleolatitude for the Hells Canyon site shown in Figure 8.

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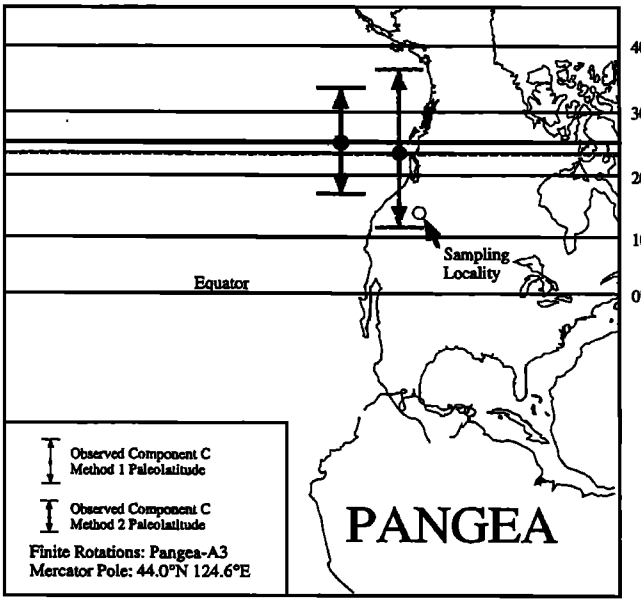


Figure 9. The mean paleolatitudes, at 95% unfolding, of magnetic component C and associated uncertainties, calculated using method 1 (only sites for which $n \geq 3$, $k \geq 10$) and method 2 (only sites for which $n \geq 3$), are shown in this reconstruction. Our mean paleolatitude suggests that the Wallowa terrane was located north of its present location on the North America plate. The paleomagnetic result, however, does not constrain the paleolongitude. The combination of our study and previous studies of the Triassic sections of the fragments of the Wrangellia terrane suggests left-lateral or sinistral transform or transpressive motion between oceanic plates in the ancient Pacific basin and the western margin of North America during Early Permian and Triassic time.

reversed magnetic polarity throughout the sampled stratigraphic section and the observation that the geomagnetic field was of reverse polarity throughout approximately 95% of the Permian to Carboniferous reversed-polarity superchron, or Kiaman interval [*Irving and Pullaiah,* 1976] imply that magnetic component C does not record a short normal-polarity event within the superchron.

Because we interpret magnetic component C as being acquired during the Long Reversed-Polarity Superchron of Early to Late Permian age we can determine a hemisphere of deposition for the Hunsaker Creek Formation and calculate a paleomagnetic pole for comparison with the averaged Early and Late Permian paleomagnetic poles and the relevant part of the APWP for North America from *Irving and Irving* [1982] (Figure 8 and Table 4).

Our interpretation of a paleomagnetic pole calculated from the component C direction compared with the APWP paleomagnetic poles of *Irving and Irving* [1982] of 268 to 232 Ma suggests that the strata of the Hunsaker Creek Formation were deposited in the northern hemisphere. The observed paleolatitude, calculated using the McFadden-Reid method for inclinations [*McFadden and Reid,* 1982] from sites of $n \geq 3$, $k \geq 10$, at optimum unfolding, is $\lambda = 24° \pm 12°$, or $26° \pm 9°$N (using $n \geq 3$ sites, at the optimum unfolding of 100%). During the appropriate age interval the expected paleolatitude for the Wallowa terrane on the North America plate during the Permian ranges between 17.0°N at 232 Ma and 13.5°N at 268 Ma using the APWP of *Irving and Irving* [1982]. This compares well with the expected paleolatitude of between 18.6°N and 14.3°N, calculated using the averaged paleomagnetic poles for Late and Early Permian calculated from reference data for North America.

Comparing observed and expected declinations shows that the Hunsaker Creek Formation records approximately 90° of counterclockwise rotation with respect to North America, using the reference APWP pole for 261 Ma from *Irving and Irving* [1982]. Because we cannot constrain the paleolongitude of this terrane, this significant counterclockwise rotation may be related to the terrane trajectory followed by the Wallowa-Seven Devils arc terrane, rotation related to collision, or postcollisional rotation.

The difference between expected and observed paleolatitude, if significant, may correspond to southward motion of the Wallowa terrane with respect to the North America plate between the early Late Permian and the Middle Triassic (about 260 to 235 Ma; Figure 9). The presence of a plate tectonic boundary between western North America and an ancient oceanic plate having a significant component of sinistral or left-lateral motion could be used to explain this displacement (Figure 9).

Summary

We have completed a paleomagnetic study of the Hunsaker Creek Formation. This formation records several components of remanent magnetization. The secondary magnetization may represent a present-day field overprint or may be interpreted to have been acquired during regional Mesozoic heating related to intrusion of plutons and suturing of the terrane to the North America plate. If interpreted as Mesozoic in age, this component has been significantly rotated in a clockwise direction since the Early Cretaceous, probably in response to distributed deformation of the edge of the North America plate related to the right-lateral transpressive motion of the Kula and/or Farallon plates with respect to the western margin of the North America plate.

We interpret the high-temperature unblocking component as having been acquired during or shortly after deposition of these rock strata in the Leonardian during the Early to Late Permian Reversed-Polarity Magnetic Superchron. This magnetic component places the Wallowa terrane in the northern Hemisphere during the Early Permian. In addition, our mean paleolatitude suggests that left-lateral motion occurred between oceanic plates in the ancient Pacific basin and western North America between the Early Permian and the Middle Triassic. This observation implies a sinistral or sinistral transpressive plate margin along the Paleozoic, western North America plate. The component C declination shows that the Hunsaker Creek Formation records significant counterclockwise rotation of the Wallowa-Seven Devils arc terrane with respect to North America.

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